The Hidden State Was Not Actually Hidden

Geist Atlas · Module 15 · How We Checked the Work

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[**CLOSED**] Tested and closed under its stated gate · **Family:** How We Checked the Work
Claim: The learner-state sensor programme closed on its current benchmark because an exact filter recovered the authored state from public history with 0.0000 bits of uncertainty at the horizon on every trajectory. There was nothing concealed for a better sensor to discover.

A zero-call reanalysis showed that the first negative result was partly a data-starved instrument, then exposed a deeper problem: the synthetic benchmark was fully observable. A future sensor test needs a genuinely ambiguous substrate.

Derived view of `paper-full-2.0.md` v3.0.298 — §6.19. The paper is canonical; edit it there, not here.

6.19 The Sensor Program Closes on a Transparent Substrate: a Two-Part Zero-Call Negative (Development-Tier)

The adaptive-state sensor program asked whether a tutor-side estimator can recover a learner's latent state from the public dialogue — the precondition for any "sense-then-adapt" architecture. Its canonical pilot had already stopped negative (lean-DAG features *worse* than a no-state baseline on both co-primary targets, −0.5212/−0.5013 log-loss nats; decision tokens `do_not_run_canonical_s2`, `validated_winner: null`, `do_not_optimize_policy`). A frozen v2.4 follow-up contract (`PLAN_4_0/2026-07-13-adaptive-state-decomposition-and-voi-protocol-v2.4.md`, committed

*This sentence is the only one actually authored by Liam Magee in this paper.

before analysis) then decomposed that stop with zero model calls, reading only checksum-verified restores of the sealed archives. It returned a two-part negative more instructive than either planned outcome.

Part 1a — the pilot null was an instrument artifact (`data_starved`). Refits reproduce the sealed stop numbers bit-exactly at the pilot's fixed penalty, and the frozen regularization sweep eliminates the negative deltas (`field_trajectory` at 16× penalty reaches the negligible band on both targets, −0.0125/+0.0146 pooled). The pilot's reference was itself weak: a schedule-only floor — predicting from (world, turn index, action family) alone — beats the pilot's no-state baseline by +0.2638 and +0.2012 nats, and that baseline carries 0.18/0.09 nats more loss than the class prior under leave-one-world-out. The stop's operative decision still stands — no rung at any penalty beats the *matched-regularization* no-state — but the failure's cause is reclassified: estimator starvation at n=144 transitions against an overfit reference, not evidence that public state lacks the signal.

Part 1b — the substrate's concealment premise was false (`close_sensor_program_on_substrate`). An exact Bayes filter over the kernel latent state (fixed-point-verified against the kernels' own oracle distributions at all 216 grid points) drives posterior entropy from 4.78–5.40 bits to 0.0000 bits at horizon *on every trajectory, under both the fixed and the greedy information-max schedules*, beating no-state by 0.66–0.84 nats. Every action family is informative (median 0.83–0.99 bits per action), so active sensing buys nothing — not because probing fails, but because nothing is hidden: the synthetic kernels, as authored, are noisy-looking but fully observable processes. The benchmark therefore never instantiated the concealed interior the program cares about; it measured estimator efficiency at small n. Any revival requires kernels whose latent state is identifiability-limited *by construction* — multiple latent trajectories consistent with every public history — which is a new instrument, not a rerun.

Status and caveats (per §5.12.6). Zero-call, sealed, and replayable end to end: report SHA-256s and the 24-dialogue/144-transition VOI arm manifest in `PLAN_4_0/2026-07-13-adaptive-state-v24-results.md` and `config/adaptive-tutor-evidence/adaptive-state-v2` verification 19/19 and 162/162 tests, deterministic replay, `model_calls: 0`. Development-tier: synthetic kernels over three detective worlds, no model in the loop, no learner claim. The operative verdicts bind: no further sensor spend on this substrate, and any learned transition model must now beat a dynamics-aware filter that is already perfect here.

Neighbouring modules: Module 7 — A Bigger Learner Model Did Not Help; Module 4 — How We Checked Every Claim.